

Byungdo Park

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RESEARCH INTERESTS

Differential geometry and algebraic topology. Secondary characteristic classes, geometric structures on higher bundles, generalized cohomology theories and their equivariant, differential, and twisted refinements, higher categories and sheaves, index theory, noncommutative geometry, and topology of matters.

ACADEMIC APPOINTMENTS

Chungbuk National University, Cheongju, Korea September 2023 – Present
Associate Professor at the Department of Mathematics Education

Chungbuk National University, Cheongju, Korea September 2019 – August 2023
Assistant Professor at the Department of Mathematics Education

Korea Institute for Advanced Study (KIAS), Seoul, Korea October 2017 – August 2019
Research Fellow (Postdoc) at School of Mathematics

The City University of New York (CUNY), New York NY, USA January 2017 – May 2017
Adjunct Assistant Professor (Hunter College and the Bronx Community College) of Mathematics

Max Planck Institute for Mathematics, Bonn, Germany October 2016 – December 2016
Postdoctoral Fellow

VISITING POSITIONS

The Hebrew University of Jerusalem, Jerusalem, Israel February 2020
Host: Zachary McGuirk

Hausdorff Research Institute for Mathematics, Bonn, Germany May 2017 – August 2017
Guest (Postdoc)
Trimester program “K-Theory and Related Fields”

Hausdorff Research Institute for Mathematics, Bonn, Germany August 2015
Guest (PhD Student)
Trimester program “Homotopy theory, manifolds, and field theories”

ADMINISTRATIVE POSITIONS

Chungbuk National University, Cheongju, Korea August 2025 – August 2027
Director of the Office of Educational Innovation
Oversees University’s Educational Innovation Center, leading comprehensive Total Quality Management (TQM) systems and evidence-based policy development for all academic programs. Directs institutional research (IR), faculty development initiatives, and smart education programs to enhance university-wide educational quality and competitiveness. Delegated Finance Officer and Delegated Contract Officer.

(Dual mandate) Chairperson of the Department of Art and Design
Oversees the Department of Fine Arts and the Department of Design. Delegated Finance Officer and Delegated Contract Officer.

(Dual mandate) Director of the Chungcheong Higher Education Compact (CHEC)
Leads collaborative educational innovation initiatives across 8 national universities in Chungcheong region, facilitating inter-university credit exchange programs and joint academic development projects. Supervises multi-institutional partnerships encompassing 50,000+ students, driving regional higher education integration and coordinating strategic educational reforms across diverse academic institutions. Delegated Finance Officer and Delegated Contract Officer.

(Dual mandate) Chairperson of the KNU10 Remote Learning Credit Exchange Council
August 2025 – February 2026
Leads national-scale remote learning collaboration framework across 10 regional hub national universities, pioneering online credit exchange programs and shared educational infrastructure. Oversees inter-institutional academic partnerships serving 200,000+ students nationwide, advancing digital education innovation and coordinating unified learning management systems across geographically distributed campuses. Drives strategic integration of high-quality online curricula, standardized academic calendars, and joint assessment protocols to expand student learning opportunities and strengthen national university network competitiveness.

	Chungbuk National University, Cheongju, Korea Associate Dean of the College of Education (Dual mandate) Director of the Teacher Appointment Support Center (Dual mandate) Director of the Future Education Center Oversees academic, research, student, planning, admission affairs of the College of Education	March 2024 – August 2024
EDUCATION	CUNY Graduate Center, New York NY, USA Doctor of Philosophy in Mathematics Thesis title: A geometric model of twisted differential K-theory Advisor: Mahmoud Zeinalian CUNY Graduate Center, New York NY, USA Master of Philosophy in Mathematics Advancement to candidacy. Yonsei University, Seoul, Korea Bachelor of Science in Mathematics Senior Thesis: Notes on L-Infinity algebras Advisor: Jae-Suk Park Member of honor program in sciences On Leave from October 2002 to February 2006 for a mandatory military service.	August 2010 – September 2016 February 2015 March 2001 – February 2009
PUBLICATIONS	On the smoothness of the geodesic spray for the Camassa-Holm equation, with Jae Min Lee, <i>J. Math. Anal. Appl.</i>, 516(1) (2022) 126472 1–12. Noncommutative differential K-theory, with Arthur Parzygnat, Corbett Redden, and Augusto Stoffel, <i>J. Geom. Phys.</i>, 174 (2022) 104446 1–40. arXiv:2106.12073. A note on Venice lemma in differential K-theory, <i>Arch. Math.</i>, 118(2) (2022) 215–224. A classification of equivariant gerbe connections with Corbett Redden, <i>Commun. Contemp. Math.</i>, 21(2) (2019) 1850001 1–40. arXiv:1709.06003. Geometric models of twisted differential K-theory I, <i>J. Homotopy and Relat. Struct.</i>, 13(1) (2018) 143–167. arXiv:1602.02292 A smooth variant of Hopkins-Singer differential K-theory, <i>New York J. Math.</i>, 23 (2017) 655–670. arXiv:1607.08686	
PREPRINTS	Brown functors of directed graphs, with Hsuan-Yi Liao, Dang Khoa Nguyen, and Zachary McGuirk, (2025) arXiv:2503.14865. Generalizations of the Thistlethwaite–Tsvietkova method, with Alice Kwon and Ying Hong Tham, (2023) arXiv:2309.01282. Geometric families define differential K-theory, with Jae Min Lee, (2024).	
REPORT	Self-Evaluation Report of the Department of Mathematics Education for the Self-Certified Major Education System of Chungbuk National University (in Korean) Principal Investigator. Certified for 4 years (1 March 2025 to 28 February 2030) Document: College of Education-14664 (2024.12.19.), Office of Education Innovation-1302 (2025.02.26.)	
WORK IN PROGRESS	Cycle maps in twisted differential K-theory, with Daniel Grady Equivariant arithmetic Chow groups and equivariant differential cohomology, with Sreedhar Bhamidi Equivariant eta forms in local index theory, with Jae Min Lee. The Euler–Arnold theory via algebraic topology: Sullivan’s methodology, with Jae Min Lee. Geometric models of twisted differential K-theory II: Non-torsion case, first draft available. Twisted Atiyah–Jänich theorem, first draft available.	
POLICY RESEARCH IN PROGRESS	A study on cycles required by the 6th teacher training institution competency assessment and the role of the college administration (in Korean) Document: College of Education-9505(2024.8.30.)	

BOOKS

Calculus (in Korean), **Precalculus** (in Korean) and **College Mathematics** (in Korean), with Daewoong Cheong, Sukjung Hwang, Un Cig Ji, Sangdon Jin, Sejong Kim, Sun Kwang Kim, Doowon Koh, Ohsang Kwon, Eunjeong Lee, Jaemin Shin, Changhun Yang, Mee Sue Yoo, Youngho Yoon, *Kyungmoon Publishing Co.*

ISBN-13: 9791160736908 (Calculus) Available to purchase at Kyobo, Aladin.

ISBN-13: 9791160736915 (Precalculus) Available to purchase at Kyobo, Aladin.

ISBN-13: 9791160736922 (College Mathematics) Available to purchase at Kyobo, Aladin.

Knowledge and education in the era of new media (in Korean), with Jin Choi, Seung-Hyun Choi, Hye-Jee Gil, Yeungyu Ha, Jiyoung Hwang, Gail Jeon, Jieun Lim, Sein Shin, Kyung Chan Son, Kyungwon Son, Jooho Song, and Hyun Wi Yoon, *Pakyoungsa*, (2022) ISBN-13: 9791165192730 Contributed a chapter “How can we be good at mathematics?”

Available to purchase at Kyobo, Aladin.

BOOK PROJECTS IN PROGRESS

Gelfand’s Geometry (in Korean) *Kyungmoon Publishing Co.*

Korean translation of the monograph *Geometry* by Israel M. Gelfand and Tatiana Alekseyevskaya (Gelfand), *Birkhäuser*, (2020) ISBN-13: 9781071602997

SELECTED PRESENTATIONS

KIAS Geometry and Topology Group Seminar

Korea Institute for Advanced Study (KIAS), Seoul, Korea

November 2024

Differential cohomology and gerbes: An introduction to higher differential geometry

Host: Seokbong Seol

Algebraic Structures in Geometry and Physics: a conference on the occasion of Jae-Suk Park’s 60th birthday

Uni Hotel Jeju, Jeju, Korea

November 2024

Aspects of differential K-theory

Host: Cheol-Hyun Cho and Hwajong Yoo

IUT Mathematics and Statistics Research Seminar (online)

Isfahan University of Technology, Isfahan, Iran

December 2023

Differential Cohomology and Gerbes: An Introduction to Higher Differential Geometry (3 lectures)

Host: Sajjad Lakzian

Mathematics Department Colloquium

Pusan National University, Busan, Korea

May 2022

An introduction to differential K-theory

Host: Donghoon Jang

Three W’s Seminar (online)

Korea Institute for Advanced Study (KIAS), Seoul, Korea

December 2020

The Atiyah-Jänich theorem and its twisted refinement

Host: Sreedhar Bhamidi

Mathematics Department Colloquium (online)

Chungbuk National University, Cheongju, Korea

November 2020

The Atiyah-Jänich theorem and its twisted refinement

Host: Sejong Kim

KIAS Topology Seminar

Korea Institute for Advanced Study (KIAS), Seoul, Korea

December 2019

Twisted differential K-theory, geometric models, and sheaves of spectra

Host: Hsuan-Yi Liao

DG Seminar

Korea Institute for Advanced Study (KIAS), Seoul, Korea

February 2019

What is an ∞ -category? (1), (2), (3)

Host: Bumsig Kim

2018 Seoul – Tokyo Conference in Mathematics

Korea Institute for Advanced Study (KIAS), Seoul, Korea

December 2018

Cycle maps in differential K-theory

Host: Jinsung Park

Topology in Australia and South Korea 2018

IBS Center for Geometry and Physics (IBS-CGP), Pohang, Korea

April 2018

A classification of equivariant gerbe connections

Host: Gabriel C. Drummond-Cole

The 13th KIAS Winter School on Differential Geometry

Korea Institute for Advanced Study (KIAS), Seoul, Korea

January 2018 – February 2018

Introduction to differential cohomology (3 lectures)

Host: Insong Choe and Jaigyoung Choe

Conference venue: Hill Condominium, High 1 Resort, Jeongseon, Korea

Higher Categories and Mirror Symmetry

Korea Institute for Advanced Study (KIAS), Seoul, Korea

December 2017

A classification of equivariant gerbe connections

Host: Bumsig Kim and Gregory Ginot

Topology Workshop – Cohomology Theories, Categories, and Applications

University of Pittsburgh, Pittsburgh, PA USA

March 2017

A geometric model of twisted differential K-theory

Host: Hisham Sati

Research Seminar

The Bronx Community College, CUNY, Bronx NY USA

February 2017

Aspects of equivariant gerbes with connection

Host: Cormac O’Sullivan

Algebraic Topology Seminar

Princeton University, Princeton NJ USA

February 2017

A geometric model of twisted differential K-theory

Host: Tony Bahri and Martin Bendersky

K-theory Seminar

Hausdorff Research Institute for Mathematics, Bonn, Germany

November 2016

Models of differential K-theory and comparison problems

Host: Kristian Jonsson Moi

Higher Differential Geometry Seminar

Max Planck Institute for Mathematics, Bonn, Germany

November 2016

A geometric model of twisted differential K-theory

Host: Christian Blohmann

Geometry and Physics (GAP) Seminar

Pennsylvania State University, University Park, PA USA

September 2016

A geometric model of twisted differential K-theory

Host: Ping Xu and Mathieu Stiénon

GAP XIV: Geometry and Physics “Séminaire itinérant” — Graded geometry and applications to physics

University of Sheffield, Sheffield, UK

August 2016

A geometric model of twisted differential K-theory

Poster presentation

Deformation Theory Seminar

University of Pennsylvania, Philadelphia PA USA

June 2016

A geometric model of twisted differential K-theory

Host: James Stasheff

2016 Lehigh University Geometry and Topology Conference

Lehigh University, Bethlehem PA USA

May 2016

A geometric model of twisted differential K-theory

GRANTS

National University Development Project – International Collaborative Research Support

Chungbuk National University

October 2024–February 2025

Project title: Construction of a Brown functor on finite directed graphs using directed graph path cohomology

Principal Investigator. Amount KRW 15,000,000 (equivalent to \$13,333)

Documents: Research Promotion Division-7193 (2024.10.22.)

Basic Research Program in Science and Engineering

The National Research Foundation of Korea (NRF)

March 2020–August 2023

The Ministry of Science and ICT, Republic of Korea

Project title: A study on differential cocycles via higher sheaves on manifolds

Principal Investigator. Amount KRW 90,000,000 (equivalent to \$80,000)

Grant No. 2020R1G1A1A01008746

National University Development Project – Creativity of next-generation researchers

Chungbuk National University

May 2022–December 2022

Project title: An approach on Euler-Arnold equations via algebraic topology: Sullivan's methodology

Principal Investigator. Amount KRW 9,000,000 (equivalent to \$8,000)

Documents: Research Support Division-3768 (2022.05.11.)

Research Grant for New Faculty

Chungbuk National University

September 2019–August 2020

Project title: Geometry and topology of directed graphs

Principal Investigator. Amount KRW 17,000,000 (equivalent to \$15,111)

Documents: Research Foundation-11099 (2019.08.28.)

Development of New Leading Scientists

KIAS School of Mathematics Group Seminar

Korea Institute for Advanced Study

January 2019–August 2019

The Ministry of Science and ICT, Republic of Korea

Project title: KIAS Topology Seminar

Co-organizer. Annual grant KRW 7,500,000 (equivalent to \$6,667)

Project title: Three W's Seminar

Co-organizer. Annual grant KRW 7,500,000 (equivalent to \$6,667)

Development of New Leading Scientists

Korea Institute for Advanced Study

October 2017–August 2019

The Ministry of Science and ICT, Republic of Korea

Project title: Differential cocycles in algebraic topology and their applications

Principal Investigator. Amount KRW 25,917,000 (equivalent to \$23,037)

Doctoral Student Research Grant

CUNY Graduate Center

June 2015–May 2016

Project title: Twisted differential K -theory, infinity categories, and mathematical physics

Principal Investigator. Amount \$1,000

CUNY Doctoral Students' Council Grant

Fall 2015

A grant for organizing seminars and conferences

Project title: The Homotopy Theory Learning Group

Principal Investigator. Amount \$700

CUNY Doctoral Students' Council Grant

Fall 2014

A grant for organizing seminars and conferences

Project title: The Algebraic Topology Student Seminar

Principal Investigator. Amount \$600

SERVICE

Scholarly Service

Referee for Journal of Geometry and Physics
Referee for Journal of the Korean Mathematical Society
Reviewer for Mathematical Reviews June 2018 – Present
Reviewer for Zentralblatt Math March 2016 – Present
Editor of Newsletter of the Korean Mathematical Society January 2023–December 2024
Reviewer for the National Research Foundation (NRF) of Republic of Korea 2024, 2023
External reviewer for tenure-track faculty hiring process June 2020 – Present
7 national universities and 13 private universities in Korea (counted with multiplicity).
External interviewer for tenure-track faculty hiring process December 2023, January 2021
2 national universities in Korea.
External reviewer for tenure evaluation process December 2025
PhD thesis reader (Committee vice chair) for 1 PhD in Mathematics Education
Organizer for Geometry Seminar at CBNU Mathematics Education dept. October 2020–Present
Organizer for Topology Seminar at KIAS January 2018–August 2019
Organizer for the three W's Seminar at KIAS January 2019–August 2019
Postdocs' seminar addressing Was, Warum, Wie (what, why, and how in German) on each research topic.
Organizer for the Homotopy Theory Learning Group at CUNY Fall 2015
Organizer for the Algebraic Topology Student Seminar at CUNY Fall 2014

University Service (at Chungbuk National University)

Ex officio positions (of the Director of Office of Educational Innovation)
September 2025 – August 2027

Member of the Academic Board
Member of the Educational Innovation Committee
Member of the Performance Management Committee
Member of the Extra-Curricular Program Management Committee
Member of the Remote Learning Management Committee
Member of the Task-Force Committee for the KNU10 Remote Learning Credit Exchange Council
Steering committee member of the Office of Educational Innovation
Steering committee member of the Office of Liberal Arts Education
Steering committee member of the Graduate School Innovation Support Program
Member of the Task-Force Committee for the Development Plan to Foster Regional Universities for Balanced National Growth (MOE project)

Ex officio positions (of the Chairperson of Department of Arts and Design)
September 2025 – August 2027

Member of the tenure-track faculty search committee Fall 2025
Member of the Research Environment Committee

Reviewer for the Executive Committee for the 6th Governmental Assessment for Teacher-training institutions
September 2024 – August 2025

Ex officio positions (of the Associate Dean of the College of Education)
March 2024 – August 2024

Member of the Planning Committee
Member of the Research Environment Committee
Member of the Admission Management Committee
Member of the Admission Management Committee for International Students
Member of the Teacher Training Committee
Member of the Undergraduate Common Curriculum Committee
Member of the Committee for Laboratory Safety Management
Member of the Committee for Energy Conservation
Member of the Curriculum committee of the College of Education
Steering committee member of the Graduate School of Education
Steering committee member of the Teacher Training Division
Steering committee member of the Education Center for Gifted-Students in Sciences
Division Head of the Administrative Support Team of the Executive Committee for the 6th Governmental Assessment for Teacher-training institutions
Associate Head of the Task-Force Committee for the University Support Project for Improving Teacher Training Programs (MOE project)

Representative of Faculty Assembly	September 2022 – August 2024
Legislation subcommittee – statute, regulations, and human resources affairs	
Member of the Institutional Assessment Research Committee	March 2024 – October 2024
Steering committee member of the Education Research Institute	December 2019 – Present
Internal admission officer	2023, 2022, 2021
Member of the Curriculum committee of the College of Education	May 2020 – May 2022
Member of the Task-Force Committee for Future Education Center construction and pilot operation project	August 2020 – February 2021

Departmental Service (at Chungbuk National University)

Member of the departmental scholarship committee	August 2020 – February 2024
Member of the curriculum committee	March 2020 – August 2023
Member of the tenure-track faculty search committee	Fall 2023, Fall 2022, Fall 2021
Member of the department officer hiring committee	Feb. 2021, Feb. 2022
Interviewer and question designer for undergraduate admission	Fall 2019 – Present
Transfer admissions (interviewer 4 times / question designer 3 times), internal transfers (interviewer 1 time / question designer 1 time), personality and aptitude interviewer (4 times).	
Interviewer and question designer for graduate admission	Spring 2023 – Present
The Graduate School of Education – (interviewer 2 time / question designer 4 times).	

Outreach and Community Service

Invited speaker for a public lecture for high school students in Chungbuk Province	
9 high school on-site, 6 high schools on-campus	July 2025– January 2026
3 high school on-site, 5 high schools on-campus	July 2024– December 2024
2 high schools on-site, 9 high schools via online	July 2023– December 2023
8 high schools on-site, 17 high schools via online (also in Sejong City)	May 2022 – August 2022
5 high Schools	December 2021
1 high School	July 2020
Review Committee Member for Shared Liberal Arts Curricular and Extra-curricular Programs, Daejeon Regional University Consortium	September 2025 – October 2025
Evaluation Committee Member for KNU10 Specialized Content Development Vendor Selection	September 2025
Advisory panel for an expansion plan of Chungbuk Mathematics Experience Center, Chungbuk Natural Sciences Education Center	November 2019 – December 2019

**TEACHING
EXPERIENCES**

At Chungbuk National University

Advanced Topics in History of Mathematics and Mathematics Education (graduate course)
Foundation of Mathematics II (graduate course)
Foundation of Mathematics I (graduate course)
Topics in Geometry I (4 semesters; graduate course)
Special Research Project II (graduate course)
Special Research Project I (graduate course)
Geometry and Education (Undergraduate topics course on teaching geometry, 2 semesters)
Differential Geometry II (7 semesters)
Differential Geometry I (7 semesters; incl. Spring 2026)
General Topology I
Complex Analysis II
Complex Analysis I
Geometry for Teachers II (2 semesters. Introduction to Euclidean, non-Euclidean, and projective geometries)
Geometry for Teachers I (7 semesters; incl. Spring 2026. Conics and the classification theorem)
Introduction to geometry (3 semesters; incl. Spring 2026)
Calculus II (2 semesters)
Calculus I (3 semesters)

At the City University of New York

Vector Analysis (Multivariable calculus and differential forms)

Calculus III (Multivariable calculus)

Calculus II (Integral calculus, 2 semesters)

Calculus II Lab (MAPLE simulation)

Calculus I (Differential calculus, 3 semesters)

Calculus I Lab (MAPLE simulation)

Precalculus

Trigonometry and College Algebra (Community College teaching experience)

College Algebra

Tasks Involved: Lecturing, grading, and writing and administering exams.

**SUPERVISION
EXPERIENCES****A study on differential cocycles via higher sheaves on manifolds** (2 students)

May 2020 – June 2021

Mentoring a undergraduate student for reading Atiyah, Bott, and Shapiro, *Clifford Modules* and a Master's student for Freed and Hopkins *Chern-Weil forms and abstract homotopy theory*.**Geometry and topology of directed graphs** (2 students)

September 2019 – February 2020

Mentored undergraduate students for learning algebraic tools for research, providing support for participating conferences, and teaching $\text{\LaTeX}$.**Characteristic Classes Learning Group** (2 students)

May 2013 – February 2014

Lecturing and mentoring in a learning group of Milnor and Stasheff's Characteristic Classes.

**TEACHING
AWARDS**

Faculty Teaching Excellence, Chungbuk National University

4 semesters: Fall 2024, Spring 2024, Fall 2021, Fall 2020, Fall 2019

**ACADEMIC
HONORS
& AWARDS**

Dissertation Year Fellowship, CUNY Graduate Center

Fall 2015 – Spring 2016

Quantitative Reasoning Fellowship, CUNY Graduate Center

Fall 2014 – Spring 2015

Graduate Teaching Fellowship, CUNY Graduate Center

Fall 2012 – Spring 2014

University Fellowship, CUNY Graduate Center

Fall 2013 – Spring 2014

Science Fellowship, CUNY Graduate Center

Fall 2010 – Spring 2012

Yonsei Special Awards, Yonsei University

Spring 2006, Fall 2006, Fall 2007, and Spring 2008

Highest Honors Student, High Honors Student, Honors Student, Yonsei University

Fall 2001, Spring 2001, and Fall 2007

University Designated Scholarship, Yonsei University

Fall 2002

The Kwanjeong Scholarship, Kwanjeong Educational Foundation

April 2002

**SELECTED
TRAVEL
SUPPORT
RECEIVED****Workshop on gauge theory and Floer homology**

December 2019

Peking University, Beijing, China.

Full accommodation support.

International Workshop on Algebraic Topology

June 2018

Southern University of Science and Technology, Shenzhen, China.

Travel award \$ 500 and full accommodation support.

Summer School "Coarse Index Theory"

September 2016

Albert-Ludwigs-Universität Freiburg, Freiburg, Germany.

Travel and accommodation award €1,580.

GAP XIV: Geometry and Physics "Séminaire itinérant" — Graded geometry and applications to physics

August 2016

University of Sheffield, Sheffield, UK

Travel and accommodation award \$1,380.

Topology of Manifolds: A conference in honour of Machael Weiss’ 60th birthday June 2016
 University of Lisbon, Lisbon, Portugal.
 Travel and accommodation award \$2,340.

GAP XIII: Geometry and Physics “Séminaire itinérant” — Derived geometry July 2015
 IBS Center for Geometry and Physics, Pohang, South Korea.
 Travel award \$1,600 and full accommodation support.

LANGUAGES **Languages** English: Fluent. Korean: Native. French and German: Basic proficiency.
Proficiency in Computer Languages C/C++, MATLAB, FORTRAN, MAPLE $\LaTeX$

SELECTED OFFERS **Korea Institute for Advanced Study (KIAS)**, Seoul, Korea March 2019
 Research Fellow (Postdoc) in School of Mathematics
 2-year extension of the contract (October 2019 – September 2021 – Resigned as of August 31, 2019).

Eduard Čech Institute for Algebra, Geometry and Physics, Prague, The Czech Republic
 16-month postdoctoral position (Offer declined) April 2017
 In the research group of Branislav Jurčo and Martin Markl

DATE Last updated: 1 March 2026.